Research Interests & Questions

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Academic status: Prospective graduate applicant. These are exploratory interests and questions, not a funded, supervised, implemented, or published research project.

BACKGROUND AND MOTIVATION

I graduated in 2025 with a BS in Information Technology from the University of Agriculture Faisalabad, Sub-Campus Burewala. Experience with programming, technical troubleshooting, and an undergraduate water-quality monitoring prototype led me to ask how computational systems handle imperfect data and unreliable evidence. I am now building my machine learning foundations and exploring directions for research-focused graduate study.

INTERESTS I WOULD LIKE TO INVESTIGATE

- Machine learning and model evaluation: how systems generalize beyond demonstration datasets and how we measure reliability.
- Edge AI and livestock behavior recognition: efficient analysis of wearable motion signals for activities such as walking, resting, feeding and rumination under computing and connectivity constraints.
- Trustworthy LLMs: hallucination, evidence-based claims, uncertainty communication and decision-making when support is missing.
- Data transparency and provenance: tracing the basis of outputs and identifying when changing or incomplete data matters.

EXAMPLE QUESTIONS ON MY MIND

- How can we recognize when an AI-generated answer is unsupported by its cited material?
- How can an efficient on-device model distinguish animal activities using small, noisy movement datasets?
- How should AI interfaces communicate uncertainty and the origin or limitations of source information?

NEXT ACADEMIC STEPS

I have not yet started formal research in these areas. My immediate priorities are to read relevant literature, strengthen my mathematics and Python/ML skills, identify realistic research questions, and seek supervision. I would be interested in fully funded research-focused Master's opportunities abroad and could pursue a PhD as a longer-term goal.

EVIDENCE OF TECHNICAL PREPARATION

AquaSense AI: exploratory Python/Google Colab notebook on water-quality data and data preparation. Portable Water-Quality Monitoring: undergraduate collection of pH, turbidity, TDS and temperature readings. ShelfAudit: AI-assisted software prototype, not a validated AI research system.

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